Development of Organic and Recyclable Waste Classification Application Using MobileNetV4 and YOLOv8

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***Abstract*—Waste management is a significant challenge, particularly in urban areas where improper waste separation reduces recycling effectiveness and increases processing costs. To address this issue, automated waste classification systems have attracted attention as a practical solution to support sustainable waste management. This study presents the development of a waste classification application that integrates MobileNetV4 and YOLOv8 as artificial intelligence models. The application is designed to classify waste into organic and recyclable categories while providing an intuitive experience through a user-friendly interface. MobileNetV4 was chosen due to its lightweight architecture and high classification performance, making it suitable for implementation on devices with limited computing resources. Meanwhile, YOLOv8 is used to provide real-time waste detection capabilities, enabling a faster and more interactive classification process. The software development process includes system analysis, application design, model integration, user interface implementation, and functional testing. The developed application allows users to upload images, receive classification results, and visualize detected waste objects through a responsive interface. The implementation results show that the integration of MobileNetV4 and YOLOv8 can provide accurate predictions, efficient processing time, and a positive user experience. The proposed application contributes to the development of smart waste management solutions by combining deep learning technologies with practical software engineering principles.**

Keywords—Waste Classification, MobileNetV4, YOLOv8, Software Engineering, User Interface, Computer Vision

I. Introduction

Waste management has become one of the most important environmental issues due to the increasing volume of waste generated by urbanization and population growth [1]. Improper waste segregation often leads to inefficient recycling processes, increased operational costs, and environmental pollution [2]. One of the key challenges in waste management is the classification of waste into appropriate categories before further processing [3].

Recent advances in artificial intelligence and computer vision have enabled the development of automated waste classification systems capable of reducing human intervention and improving sorting efficiency [4]. Deep learning models have demonstrated promising results in image recognition tasks and have been widely adopted in environmental monitoring applications [5].

In previous research, various deep learning architectures were evaluated for waste classification tasks. Among these models, MobileNetV4 and YOLOv8 achieved superior performance in terms of classification accuracy, computational efficiency, and real-time processing capability [6]. These results indicate that both models have strong potential for practical implementation in intelligent waste management systems.

From a software engineering perspective, the successful deployment of artificial intelligence models requires not only accurate predictions but also an effective system design and user-friendly interface [7]. User experience plays a crucial role in ensuring that the application can be utilized effectively by both technical and non-technical users [8].

Therefore, this study focuses on the development of a waste classification application that integrates MobileNetV4 and YOLOv8 into a single platform. The proposed system aims to provide accurate waste classification, real-time detection capability, and an intuitive user interface to support practical waste management activities [9]. Furthermore, the application is designed to facilitate practical implementation in smart waste management environments while maintaining efficiency and usability [10].

II. Literature Review

Recent studies have demonstrated the effectiveness of deep learning techniques in waste classification applications. Gupta et al. [11] proposed an AI-based waste segregation system that improved classification efficiency and reduced manual sorting efforts. Similarly, Zhao et al. [12] developed an intelligent waste management framework by integrating artificial intelligence and smart technologies.

MobileNet-based architectures have attracted significant attention due to their lightweight design and suitability for resource-constrained devices. Rahman et al. [13] reported that MobileNet achieved reliable performance for real-time waste classification while maintaining low computational requirements.

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In addition, YOLO-based models have become widely adopted for real-time object detection tasks. Kumar et al. [14] demonstrated that YOLO can accurately detect waste objects under different environmental conditions. The latest YOLOv8 architecture further improves detection accuracy and inference speed, making it suitable for practical deployment [15].

Based on these findings, the combination of MobileNetV4 and YOLOv8 offers a promising solution for developing an intelligent waste classification application that balances accuracy, efficiency, and usability.

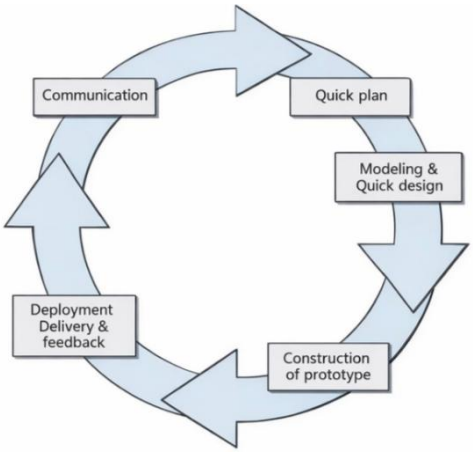
III. System Design & Methodology

A. System Requirement Analysis

The requirements analysis phase was conducted to identify the needs and expectations associated with the development of the EcoGreen application. This phase aimed to determine the functionalities required to support intelligent waste management activities while ensuring that the application remains practical and accessible to users. The analysis revealed that the application should be capable of allowing users to upload waste images from their device gallery or capture images directly through the smartphone camera for classification purposes. The system is expected to classify waste into organic and recyclable categories using the MobileNetV4 model and provide real-time object detection using YOLOv8. In addition to classification capabilities, the application should provide educational information regarding waste categories through the Trash Encyclopedia module, assist users in locating nearby waste banks through the Waste Map feature, and encourage sustainable behavior through the EcoBadges module that records achievements and environmental contributions.

Apart from functional requirements, several non-functional requirements were also identified. The application should provide an intuitive and user-friendly interface to ensure ease of use among users with varying levels of technical experience. Since the proposed system targets mobile environments, the application should maintain efficient processing performance and deliver prediction results within an acceptable response time. Furthermore, the system architecture should support maintainability and future expansion, allowing updates to the integrated artificial intelligence models without requiring substantial modifications to other application components.

B. Development Process (Prototyping Process Model)



Workflow of the proposed waste classification system, including data preprocessing, model training, evaluation, and model deployment.

The software development methodology adopted in this study is the Prototyping Model, a process model that emphasizes iterative development through the construction and refinement of working prototypes based on continuous user feedback. Unlike waterfall-based approaches, the Prototyping Model is particularly well-suited for systems where user requirements are not fully defined at the outset, as is common in user-facing mobile applications such as EcoGreen. The process, as illustrated in Fig. 1

- **Communication:** The first phase involves gathering initial requirements from stakeholders and end users. In the context of this project, this phase captured the functional needs of waste classification, including image upload capability, real-time detection, and educational waste information. The goal of this phase is to establish a shared understanding between developers and users before any design work begins.
- **Quick Plan:** Following communication, a rapid planning session is conducted to define the scope, timeline, and resources for the current prototype iteration. This phase does not involve detailed project planning; instead, it produces a lightweight plan sufficient to guide the immediate design and development cycle.
- **Modeling & Quick Design:** In this phase, developers produce an initial system design including application architecture, data flow, and user interface mockups. The design is intentionally simplified to enable fast iteration. For EcoGreen, this phase produced wireframes for the Smart Scanner, Trash Encyclopedia, Waste Map, and EcoBadges modules, as well as the three-layer architecture connecting the presentation, application, and model layers.
- **Construction of Prototype:** The prototype is built based on the quick design. This phase involves actual coding and integration of components, including model deployment via REST API, frontend implementation in Flutter, and backend configuration in Flask. The resulting prototype is a functional but preliminary version of the application, sufficient for user evaluation.

- **Deployment, Delivery & Feedback:** The prototype is delivered to users or testers for evaluation. Feedback is collected regarding usability, correctness, and missing functionality. This feedback directly informs the next iteration of the Communication phase, restarting the cycle. The process repeats until the prototype is refined into a final, approved system.

The adoption of the Prototyping Model allows the EcoGreen application to evolve iteratively according to user needs and technical constraints identified during development. Through continuous feedback and refinement cycles, the resulting system is expected to provide both functional accuracy in waste classification and satisfactory user experience in promoting environmental awareness.

IV. Implementation

A. *Smart Scanner Module*

The Smart Scanner module serves as the core feature of the EcoGreen application. This module enables users to upload images or capture waste objects directly using a camera for classification. Once an image is submitted, the application performs preprocessing and sends the image to the integrated MobileNetV4 model for classification. The prediction results are then displayed through the user interface, including the waste category and confidence score.

To improve usability, the interface provides a simple upload mechanism, visual feedback during processing, and clear classification results. The lightweight architecture of MobileNetV4 allows the system to deliver fast predictions while maintaining low computational requirements, making the application suitable for deployment on various devices.

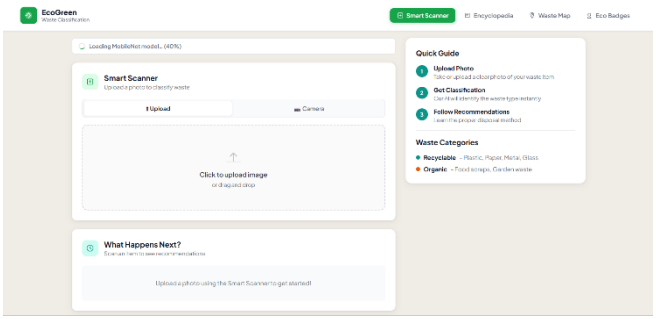


Fig. 1. Smart Scanner interface

Figure 2 presents the Smart Scanner interface, which allows users to upload or capture waste images. The system processes the image using MobileNetV4 and displays the predicted waste category through an intuitive user interface.

B. *Trash Encyclopedia Module*

The Trash Encyclopedia module was developed to provide educational information regarding waste classification and proper disposal methods. This feature allows users to search for different waste items and access detailed information about their categories, characteristics, and recycling recommendations. By combining waste classification functionality with educational content, the application aims to increase user awareness of sustainable waste management practices.

The interface was designed to support quick information retrieval through a searchable waste database and visual category indicators. This module complements the classification system by helping users understand the reasoning behind waste categorization and encouraging environmentally responsible behavior.

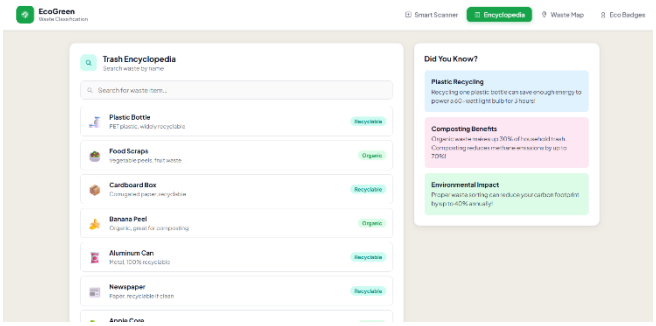


Fig. 2. Trash Encyclopedia interface

Figure 3 shows the Trash Encyclopedia module that provides educational information regarding different waste categories. Users can search waste items and obtain disposal recommendations and environmental information.

C. *Waste Map Module*

The Waste Map module assists users in locating nearby waste banks and recycling facilities. This feature was designed to bridge the gap between waste identification and proper disposal by providing accessible information about waste management locations. Users can view nearby facilities through an interactive map and obtain location details to support recycling activities.

The interface presents waste bank information in a user-friendly format, including facility names, distances, and addresses. This functionality encourages active community participation in waste management by simplifying access to recycling infrastructure.

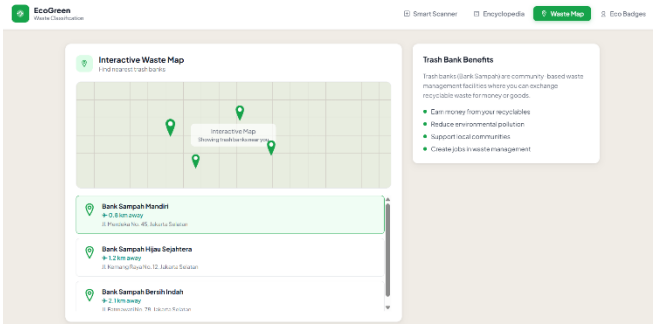


Fig. 3. Waste Map interface

Figure 4 illustrates the Waste Map feature, which helps users locate nearby waste banks and recycling facilities. This functionality supports proper waste disposal and recycling activities.

D. *Eco Badges Module*

The EcoBadges module was developed to enhance user engagement through a gamification-based approach. This feature records user activities related to waste classification and presents them in the form of achievements, badges, and

progress statistics. Users can monitor the total number of waste items successfully classified, track their environmental contributions, and unlock achievement badges based on specific milestones.

In addition to personal progress tracking, the module provides a social sharing feature that allows users to share their achievements on social media platforms. This functionality aims to encourage continuous participation in waste sorting activities while promoting environmental awareness within the community. By integrating gamification elements into the application, EcoBadges improves user motivation and supports long-term engagement with the waste management system.

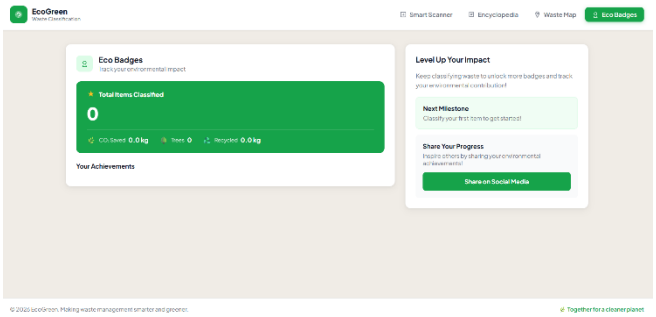


Fig. 4. EcoBadges interface displaying user achievements and classification progress.

Figure 5 shows the EcoBadges module, which provides achievement badges and progress statistics based on users' waste classification activities. Users can track the total number of successfully classified items and share their achievements on social media. This gamification feature is designed to increase user engagement, encourage consistent waste sorting behavior, and promote environmental awareness through an interactive and rewarding user experience.

V. Result and Discussion

The MobileNetV4 model was successfully integrated into the EcoGreen application as the primary image classification component. The model was able to classify waste images into organic and recyclable categories with high prediction accuracy and fast response time. Due to its lightweight architecture, MobileNetV4 required relatively low computational resources, enabling efficient deployment within the application environment.

The classification results were displayed directly through the Smart Scanner interface, allowing users to obtain waste category information in a simple and intuitive manner. During testing, the model consistently generated predictions with minimal processing delay, providing a smooth user experience. These results demonstrate that MobileNetV4 is suitable for practical waste classification applications, particularly on devices with limited hardware capabilities.

YOLOv8 was integrated into the system to provide real-time waste object detection functionality. The model successfully identified waste objects and generated bounding boxes around detected items while maintaining stable detection performance. The implementation enabled users to visualize detected waste objects and receive classification information simultaneously.

The real-time detection capability of YOLOv8 enhanced the interactivity of the application by allowing users to observe waste recognition results instantly. During system testing, the model demonstrated efficient inference performance and maintained reliable detection results under various image conditions. The integration results indicate that YOLOv8 can effectively support intelligent waste monitoring and classification applications.

The EcoGreen application was developed using a user-centered design approach to ensure ease of use and accessibility. The implementation of the Smart Scanner, Trash Encyclopedia, Waste Map, and EcoBadges modules provided users with a complete waste management experience beyond simple waste classification. The interface design emphasizes clear navigation, responsive interaction, and informative visual feedback.

The Smart Scanner module allows users to classify waste images quickly, while the Trash Encyclopedia provides educational information regarding waste categories and disposal methods. The Waste Map feature assists users in locating nearby waste banks and recycling facilities. Furthermore, the EcoBadges module introduces gamification elements through achievement tracking and social sharing features, encouraging continuous participation in waste sorting activities.

Overall, the integration of artificial intelligence models with user-friendly interface components resulted in a functional and engaging application. The combination of accurate classification, real-time detection, educational content, and gamification features contributes to a positive user experience and supports sustainable waste management practices.

VI. Conclusion

This study successfully developed the EcoGreen waste classification application by integrating MobileNetV4 and YOLOv8 into a single intelligent waste management platform. MobileNetV4 was implemented as the primary image classification model due to its lightweight architecture and efficient performance, while YOLOv8 provided real-time object detection capabilities that enhanced the system's functionality and interactivity.

The application was designed using a user-centered approach and includes several key features, namely Smart Scanner, Trash Encyclopedia, Waste Map, and EcoBadges. These features not only support waste classification but also provide educational content, location-based services, and gamification elements to improve user engagement and environmental awareness.

The implementation results demonstrate that the integration of deep learning models with a well-designed user interface can produce an effective and practical waste management solution. The developed system is capable of delivering accurate classification results, efficient processing performance, and a positive user experience. Future work may focus on expanding waste categories, improving detection accuracy, and deploying the application on mobile platforms to increase accessibility and real-world adoption.

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DATA AVAILABILITY

The dataset utilized in this research is publicly accessible through Roboflow and is available at: https://universe.roboflow.com/shierens-workspace/trash-data-classification. The dataset contains annotated waste images categorized into organic and recyclable classes, which were used for model training, validation, and testing.

All data preprocessing, model integration, application development, and system evaluation were conducted using Google Colab and related software development tools. Additional materials, including implementation details and source code, can be obtained from the corresponding author upon reasonable request to support transparency and reproducibility of the research findings.

AUTHOR CONTRIBUTION

Hans Christian Kosasih was responsible for the system design, application development, and integration of the deep learning models. Shieren Revalina Jesslyn contributed to the preparation, review, and refinement of the manuscript. Virlene Marchella conducted data collection, dataset preparation, and system testing activities. Yasi Dani provided research supervision, technical guidance, and academic feedback throughout the study. All authors contributed to the discussion of the results and approved the final version of the manuscript.